Decoding the Emotional Voice: A Searchlight fMRI Analysis of Nonverbal Vocalizations

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Abstract

Communicating emotions is a crucial part of human interaction. This communication can be done through nonverbal vocalizations such as laughing or moaning.

This study analyzed the functional magnetic resonance imagery (fMRI) data collected while participants listened to others' nonverbal emotion vocalizations.

A sliding local neighborhood multivariate analysis was applied to the fMRI data of five participants to answer the questions whether it is possible to decode the five emotions from fMRI data during listening to nonverbal emotional vocalizations and which areas are most involved in this process. Using this analysis, one map of decoding accuracies was calculated per participant. These were then averaged to find common high accuracy areas across the participants.

The resulting map shows that models were able to successfully distinguish between the five emotions with an accuracy of up to 0.29 (chance level: 0.2). Clusters with above 0.25 accuracy were found in the left and right STC, right IFG, right supra-marginal and angular, left and right SFMC, precuneus as well as the left pre- and postcentral gyri.

These results support existing theories about the processing of emotional auditory cues in human communication and extend a broad body of research by focussing on the less studied mode of communication of nonverbal vocalizations.

1 Introduction

The ability to understand others emotions is a crucial part of any social interaction. The consequences of losing this ability, say through certain neuro-developmental or degenerative disorders, are severe, affecting emotional empathy, social functioning and communication as a whole. Despite the severity, the understanding of the neural foundation of emotions even in healthy brains remains incomplete. Treatment and care of patients with empathy-impairing disorders, such as autism spectrum disorders or certain types of dementia (see Bartochowski et al. (2018)), could heavily benefit from better understanding of the neural representation of empathy.

The way humans communicate emotions can be divided into verbal (through words and speech prosody) and nonverbal communication. Nonverbal communication ranges from body language to facial expressions to nonverbal vocal expressions such as laughing (to express *happiness*) or screaming (for *anger*). A key motivation to study nonverbal vocal expressions, is that certain nonverbal emotional vocalizations seem to be understood across cultures (Sauter et al., 2010), which suggests universal acoustic cues that allow humans to communicate emotions across languages. This study analyzes the neural responses of participants listening to emotional nonverbal vocal expressions.

The proposal of a network of neural regions responsible for emotional processing by Papez (1937) marked a milestone in the neural research of emotions and was followed by an increased interest in the study of this field. With modern technologies, such as magnetic resonance imaging (MRI) and electroencephalography (EEG), and better methods, current research found emotional processing of emotional auditory stimuli to be constituted by bidirectional connections of cortical and subcortical brain regions. Brain areas often found to be involved in this process are the superior temporal cortex (STC) (specifically the auditory cortex (AC)), amygdala, inferior frontal gyrus (IFG), insula and medial frontal cortex (MFC) (Frühholz et al. (2016), Liu et al. (2023), Ethofer

et al. (2009)). This bidirection interplay can be divided into two stages, where the first stage is a quick initial response that is then refined in the second stage. The first stage of this process takes place in the auditory cortex, which already determines the emotional content of the incoming signal and decodes an initial vocal emotion (Schirmer and Kotz (2006), Ethofer et al. (2012)). In the second stage, this information is passed on to the amygdala which seems to play the role of a hub and center of communication to many other cortical areas and networks for further integration and interpretation of the stimulus as well as the regulation of the emotional response (Schirmer and Kotz (2006), Pessoa (2017), Sachs et al. (2018)).

In order to stimulate certain emotional reactions in the participants' brain, many studies rely on the phenomenon of affectional empathy, which refers to the process that allows humans to feel someone else's emotions just by observation and perception of the other. Through this, previous research was able to successfully decode emotions from fMRI images of participants perceiving others emotions (Wallenwein et al. (2024), Kotz et al. (2013)).

The decoding of emotions from fMRI images is a field of study that finds a lot of attention in current research and has been studied across different affectional stimuli. The stimuli with which the participants were faced can be divided into visual stimuli (Wallenwein et al. (2024), Fusar-Poli et al. (2009)) and auditory stimuli. The auditory stimuli ranged from affectional music (Sachs et al., 2018) to emotional intonation of non-words (Kotz et al., 2013). However, nonverbal emotional vocalizations remained (to the best knowledge of the author) an underexplored stimulus in the context of decoding emotions from fMRI. This study fills that gap by applying a sliding local neighborhood multivariate analysis, called searchlight analysis, to decode the five emotions from fMRI collected during listening to others nonverbal emotional vocalizations: *Anger*, *Relief*, *Disgust*, *Awe* and *Neutral*. The research questions this study answers through this analysis are:

1) *Is it possible to decode emotions from fMRI images taken during listening to nonverbal emotional vocalizations?*

2) *Which areas are most involved in the emotional processing of the nonverbal emotional vocalizations?*

2 Material and Methods

2.1 Data

The fMRI images this study analyzed have been collected in the Stockholm University Brain Imaging Center (SUBIC) (section 7 for more information). The experimental design consisted of nine runs, where each run consisted of 40 blocks, where there was four blocks for each task-emotion pair. This resulted in 180 fMRI images per participant per task. The tasks were to either express or to listen to a nonverbal vocal expression of one of the five emotions: *anger*, *awe*, *disgust*, *relief* and *neutral*. After having listened to or expressed an emotions, participants had to give a rating on a rating scale from 1 to 3 of how strongly they felt the emotion. During each run the blood oxygen level dependent (BOLD) signal has been recorded with a precision of 2x2x2 mm. The average time of one run was about 10 minutes.

The raw BOLD images have been realigned to correct for motion, then normalized into a MNI space by using an anatomical T1 weighted image of the same subject, then smoothed (6x6x6 mm Gaussian FWHM). Furthermore, outlier detection and artifact removal was applied due to the motion-involving task to express vocalizations. Normalization and smoothing have been done using the software package for statistical parametric mapping SPM, outlier detection as well as artifact removal have been done using the ArtRepair Toolbox for SPM.

Next, β-images were extracted by fitting a general linear model to the fMRI data. β-images are statistical maps, where for each voxel in the fMRI image and for a set of predictors (such as task conditions or movement in the scanner) in the design matrix, a separate beta value is calculated, which represents the BOLD signal strength in that voxel associated with that predictor. The general linear model was fitted using SPM and contained predictors for each experiment block, for all rest periods, all rating periods as well as for movement. This analysis is done in order to separate noise and BOLD-activations that are independent from (or, at least, not clearly correlating to) the presented stimulus (for a critical review of this technique, see (Monti, 2011)).

In this study only the β-images corresponding to the listening task blocks have been used for the analysis. The listening task has been selected due to having less movement-related noise in the data.

Furthermore, should the analysis yield positive answers to the research questions, this passive listening task still allowing for decoding, supports theories of empathy and the humans ability to share others' emotion by just perceiving their affective sound cues.

2.2 Searchlight Analysis

Analysis has been conducted using searchlight analysis. Searchlight analysis is a form of multivariate pattern analysis (MVPA), in which an MVPA accuracy score is calculated for each voxel, using the voxel and its neighbors within a certain predefined radius. This process can be thought of as moving a sphere through all voxels in the brain and training a model after each step on only the voxels inside this sphere, hence the name *searchlight*. The accuracy of each model is stored in the voxel that is at the center of that sphere. This map of local accuracies (also simply called accuracy-map) allows for precise localization of brain areas whose activations are able to distinguish between the different stimuli. Despite shortcomings of this analysis method, such as potentially missing larger distributed patterns (see (Etzel et al., 2013) for a critical review of searchlight analysis), searchlight analysis is a widely used method to study involvement and decoding-potential of local neighborhoods of voxels and has already been successfully applied to predict emotional categories (Sachs et al. (2018), Kotz et al. (2013), Wallenwein et al. (2024)).

To decode the five emotions used in this study, a searchlight model was trained for each of the five participants fMRI data separately. A searchlight sphere with radius of 12x12x12mm (6x6x6 voxels, on average 682 (707 median) voxels per model) has been used to calculate the accuracy maps for each of the participants. A linear support vector classifier (specifically scikit-learn's implementation) was trained on each sphere using leave-one-run-out cross validation to avoid leakage of temporally correlated within-run data between the training- and validation sets. The five resulting accuracy maps were then averaged to obtain a general accuracy map. It is, therefore, an average accuracy map of within-participant accuracy results. The resulting maps did not undergo statistical significance testing, due to lack of resources (see section 4 for more information).

The searchlight analysis for this study has been implemented using the *Nilearn* python package (see nilearn.github.io for more information).

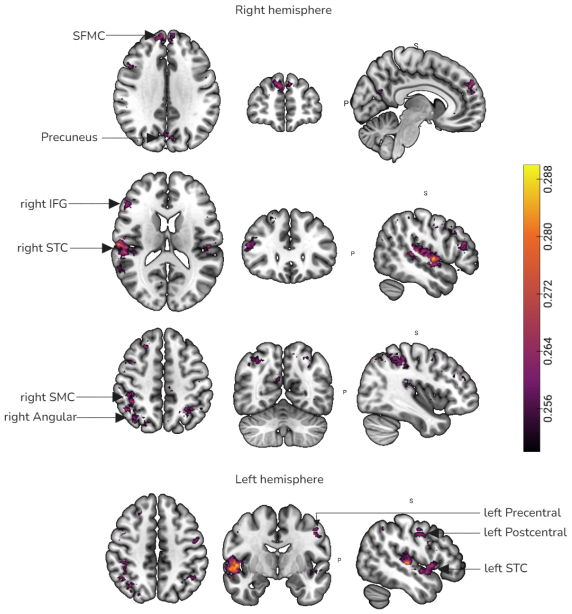

Figure 1: Accuracy map showing the results of the searchlight analysis. This accuracy map shows for each voxel the average accuracy across all participants. Map has a lower bound threshold of 0.25, in order to attain the best image clarity as well as to show only accuracies well above chance. Abbreviations used in this figure: SFMC - Superior Frontal Medial Cortex, IFG - Inferior Frontal Gyrus, STC - Superior Temporal Cortex, SMC - Supra-Marginal Cortex

3 Results

The mean accuracy map is the average accuracy across the five within participant maps. Even in this averaged map there are regions with accuracies of up to 0.29, which is well above the chance level of 0.2 (see Figure 1). The main regions with exceptional decoding performance of at least 0.25 are superior temporal cortex (STC) (bilateral), precuneus, superior frontal medial cortex (SFMC), right inferior frontal gyrus (IFG), supra-marginal gyrus (bilateral), left angular gyrus as well as left pre- and postcentral gyri. The aforementioned maximum of 0.29 is achieved in some single voxels populated in both the left and right superior temporal gyri as well as the left frontal superior medial cortex (see Figure 2).

4 Discussion

Both research questions have been answered positively by the results. There was a multitude of areas with above chance accuracy, and there are

clear outstanding areas in which the searchlight analysis was able to decode the emotional categories. The results of this analysis serve as further evidence that there are fine-grained spatial neuronal patterns that contain distinctive information to allow for decoding of the emotion expressed through nonverbal vocal expressions. The accuracy map shows increased (and above chance) decoding performance in regions that align with previous studies. As such the superior temporal cortex has been found to form distinctive patterns that allow for decoding of emotion from vocal features and other auditory stimuli (Sachs et al. (2018), Ethofer et al. (2009), Frühholz and Grandjean (2013)). This suggests that the role of the STC in auditory processing, includes to already encode information from the stimulus, which align with previous work suggesting an early encoding of information from the auditory stimulus which is then passed on for further processing (Ethofer et al., 2012). The surprisingly low decoding accuracy for voxels in the amygdala (decoding accuracy below chance in both hemispheres) may be linked to an emotion-unspecific involvement during emotional processing of the emotions picked for this study. Speaking of surprise, increased decoding in the right inferior frontal gyrus (IFG) converges with findings in (Kotz et al., 2013), where there was increased IFG decoding accuracy, especially for the *surprise* emotion. The activation patterns of *suprise* have previously been linked to patterns during situations where the outcome (or 'reward') is uncertain, which in turn has been found to lead to increased activation in selective-attention related areas such as the precuneus (Chandrasekhar et al., 2008). This supports the result found in this study, where neighborhoods in the precuneus were able to decode emotions with accuracies of around 0.26. The areas of high decoding accuracy superior frontal medial cortex (SFMC), precuneus and the angular gyrus are all part of the default mode network (DMN). The DMN plays an important role in a multitude of tasks, including mentalizing the state of mind of others as well as emotion-related retrieval and tracking of autobiographical memory (Andrews-Hanna (2012), Menon (2023)). This study gives further support for the involvement of the DMN during emotional encoding, as there evidently exist patterns that allow for emotion-prediction in regions belonging to the DMN. The involvement in the encoding of emotions in the precentral gyrus, which encompasses motor regions, could assist the mirror neuron theory (Dimberg et al. (2000), Corradini and Antonietti (2013)). Mirror neurons are a type of neuron that is thought to directly mirror (or mimic) the bodily movements or emotions of others while merely observing them. This mirror neuron system, whose existence is to this date still debated, is also said to play an import role in empathy (Iacoboni, 2009). The increased decoding performance of both the precentral gyrus, which contains motor regions, and the postcentral gyrus, containing somatosensory regions, might be connected to the participants mimicking of the other persons movements in the vocal tract and the mirroring of their mental state.

Despite the strong decoding performance found through the searchlight analysis performed in this study, there are ways to improve the power and generality of the found results. Firstly the results presented here are raw accuracy maps. That is, due to limited time, a nonparametric statistical test was impossible and, furthermore, the insufficient amount of participants lead to poor results in the parametric statistical tests. It is, therefore, important to repeat the analysis performed in this study, once there is more data available and to back those future analyses with statistical tests. Secondly, it is important to highlight the limitations of searchlight analysis. Due to the spherical shape and limited size of each searchlight model, it is possible to miss patterns that are either more spatially distributed in size or shape. For a more in-depth analysis of the advantages and disadvantages of searchlight analysis, the author recommends (Etzel et al., 2013). Future research can further extend this study by analyzing emotion specificity of these areas. In this study merely decoding accuracy was looked at. However, analyzing per-emotion or emotion-category (such as positive vs. negative emotions) could lead to more insights into the specific involvement of areas during the processing of the given stimulus or result in finding tendencies of areas to be more activated during processing a specific set of emotions (during auditory stimulus processing).

5 Ethics

The study whose data was analyzed here was followed the guidelines and was approved by the Swedish Ethical Review Authority (Etikprovn-

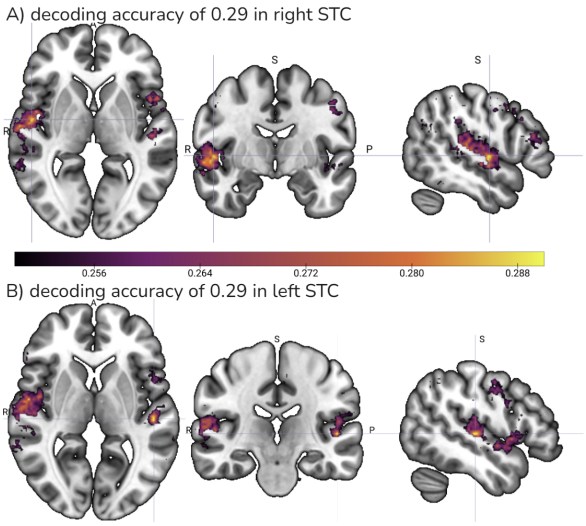

Figure 2: Regions that attained maximum decoding accuracy through the searchlight analysis, which was found in both the left and right superior temporal cortex

ingsmyndigheten). All participants have been informed about the study and gave informed consent. The data collection was made respectful of the participants comfort and was not dangerous. The participants have been able to stop the experiment at any point without giving a reason. The collected data has been stored so as to not be able to be tracked back to the participants identity. The participants also consent to forwarding taken images to a radiologist in case there should be any incidental findings. As for the ethical considerations of the second level analysis performed in this study, it should be noted that the accuracy maps are no direct evidence for emotional representations. The classifier has merely predicted one of the five emotional categories used as stimulus. This limited scope is important to mention in order to avoid over-interpretation such as thinking that this model was able to 'read the persons mind'. Furthermore, this study does not claim that the results found in this study are demographically independent. The participants for this study were Swedish speaking adults.

6 Conclusion

This study applied searchlight analysis to find fine-grained local neighborhoods in fMRI images that play a role in the encoding of emotions during perception of emotional nonverbal vocalizations. The resulting accuracy maps contained areas with decoding accuracy of up to 0.29 (chance level: 0.2) with clear clusters forming when (lower bound) thresholding at 0.25. These clusters were located in the left and right STC, right IFG, right supramarginal and angular, left and right SFMC, precuneus as well as the left pre- and postcentral gyri. The results align with previous research that suggests that emotional encoding already happens in the primary auditory cortexs located in the STCs. Interestingly, the results also show involvement of most parts of the DMN. The increased decoding accuracy in the pre- and postcentral gyri are in support of the mirror neuron theory, as they might suggest the brains mimicking of the physical movements involved in the production of the stimulus vocalizations. This study supports the current understanding of processing of emotional auditory stimuli and extends it by showing areas that are relevant for emotional en- and decoding in the human brain. However, further research is needed to fully understand how our brain grasps the hard to grasp things that are emotions.

7 Acknowledgments

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